

Software Project Management Plan

Budget Bridge Web Application

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1 Overview

1.1 Purpose and Scope

This document provides an overview of the circumstances and the course of action we will take when it comes to building out the Budget Bridge web application, which is described in our project charter document. Describing the circumstances of the Budget Bridge web application involves describing details that give the reader context about what's expected to be included in Budget Bridge and how the developer team is going to go about building and delivering a working version of the Budget Bridge web application.

1.2 Assumptions, Constraints, Dependencies and Resources

Assumptions are conditions that the developer team of Budget Bridge depend on to be true for the entirety of the time spent working on the project in order to be successful, but they are outside of the developer team's control.

Constraints are limitations that are placed on the Budget Bridge. The constraints come from our professor for University of Missouri—Kansas City's (UMKC) Software Engineering Capstone (CS-451R) course and the Commerce Bank team.

Dependencies are relationships between tasks that are predecessors or successors of each other. These define the order in which tasks are done.

Resources are things that the developer team will need to use in order to successfully complete this project.

Assumptions:

1. All developers will contribute adequate effort to complete project tasks
2. The frontend and backend technologies that developer team picked at the beginning of the project need to be able to integrate with no issues
3. At the end of the project, the developer team needs to be able to showcase a full working demo of the Budget Bridge web application.
4. All developers have access to good enough hardware (computers) that they can use in order to work with the technologies that are used to make Budget Bridge

Constraints:

1. A CMS (like WordPress) cannot be used to create the web application

Dependencies:

1. The API that is used to connects the frontend and the backend needs to be able to establish the connection with no issue, because the frontend cannot function without the backend
2. The content that's shown in the UI/frontend is dependent on the backend being able to send the data from the MySQL database back to the frontend without any issue.
3. Being able to update the MySQL depends on the frontend being able to successfully send API request to the backend
4. The data stored in the MySQL database needs to accessible to the backend

Resources (Technology and non-technology resources):

1. JIRA: The project management software that will be used to track tasks
2. GitHub: The version control system that will be used for code collaboration
3. Microsoft Teams: The communication platform that will be used for communication between developers
4. Calendar Days: The developer team will approximately have

1.3 Timeline (tasks and milestones)

This section provides an overview of the key deadlines and milestones and what tasks and other things are associated with each deadline and milestone. This shows how the Budget Bridge web application project is progressing over time.

Milestone #1 on Mar 12, 2026: First in-person demo of Budget Bridge

By Mar 12, 2026, the following items will be delivered:

1. A basic working version of the Budget Bridge application that has the frontend and backend connection working and that successfully demonstrates 2 functionalities, which are registering and logging in and viewing the user's data in the dashboard page.

Milestone #2 on Mar 30 ,2026: First app demo with the Commerce Bank team over Zoom

By Mar 30, 2026, the following items will be delivered:

1. A more frontend polished working version of the Budget Bridge application to be demonstrated over zoom

By May 1, 2026, the following items will be delivered:

1. All 8 required documentation documents, as described in the second main project deliverable, are completed and published online, ready to be viewed.

By May 8, 2026, the following items will be delivered:

1. A final working product of the Budget Bridge web application, ready to be demonstrated over the Zoom meeting with the Commerce Bank team that incorporates the feedback given by the Commerce Bank team from Mar 30's demonstration over Zoom.

1.4 Project Deliverables

This section lists all of the deliverables that will be delivered by the developer team at the end of the project.

Deliverables to be delivered by May 4, 2026:

1. Source code for the frontend and backend of the application
2. 8 Documentation Documents:
 1. Project Charter
 2. Requirement Document
 3. Project Plan
 4. Architecture/Design Document
 5. Test Plan
 6. Test Specification (Test Cases)
 7. Test Report
 8. Team Effort Estimation
 9. User Guide
3. Final Project Presentation: This presentation will provide an overview of the Budget Bridge web application from the project management perspective and a final demo of the working web application will be shown that demonstrates all of the core functionalities.